HUMAN FOLLOWING ROBOT

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Abstract — *Human following robot is a robot system which will detect and analyze human gestures and respond to them accordingly. It will also be able to interact with its surrounding environment. The robot system consists of sensors, motor drivers, arduino UNO, DC motors etc.*

Keywords — Human following*, *human tracking*,** *visual imaging*, ***human robot *interaction, laser range sensors*

I. INTRODUCTION

Robotic technology has increased appreciably in the past couple of years. Such innovations were only a dream for some people a few years back. But in this rapid moving world, now there is a need of robots such as “A Human Following Robot” that can interact and co-exist with them. Tasks performed by humans have its own limitations in many ways. In order to perceive beyond the human limitation in vision, speed, consistency, flexibility, quality etc we need robots. The problem statement focused on is creating an efficient and portable mobile robot system which can operate on human commands. The main aim of this project is to provide a blueprint for larger projects which require same basic functionality of robot.

There are many interesting applications of this research in different fields whether military or medical. A wireless communication functionality can be added in the robot to make it more versatile and control it from a large distance. This capability of a robot could also be used for military purposes. By mounting a real time video recorder on top of the camera, we can monitor the surroundings by just sitting in our rooms. We can also add some modifications in the algorithm and the structure as well to fit it for any other purpose. Similarly it can assist the public in shopping malls. So there it can act as a luggage carrier, hence no need to carry up the weights or to pull that. Similarly, ample amount of modifications could be done to this prototype for far and wide applications.

II. LITERATURE REVIEW

Study of ultrasonic sensors in human following robot

This paper discusses merits of ultrasonic sensors in robot, which is used for human presence detection. In this robot 3 ultrasonic sensors of 40 KHz frequency are used for detection and the robot technology has been tested in many different angles of the sensors. Many of the existing face recognition techniques work efficiently on frontal faces of similar sizes with proper illumination conditions. However, this assumption may not hold in real-life scenario, due to the varied nature of face appearance and environment conditions. The extraction of faces is necessary for reliable face classification techniques.

Human following robot using Arduino UNO

In this paper the robot has Infrared sensors used to move it in both the direction and ultrasonic sensor for both the forward and reverse direction. We used the Arduino Uno microcontroller as the brain of this project. This robot is driven with four Dc motors and it is controlled by a motor driver shield.

Design and Development of Human Following

In this paper, we presented a method of a human following robot based on tag identification and detection by using a camera. Intelligent tracking of specified target is carried out by the use of different sensors and modules ultrasonic sensor, magnetometer, infrared sensors and camera.

Study of vision and object tracking in human following robot using ultrasonic and infrared sensors

This paper discusses about the success and problems encountered during vision and object tracking in a human following robot. It specifically tells about the algorithms, sensors, circuit arrangement, sensor calibration and microcontroller operations.

III.METHODOLOGY

Components

The robot system uses sensors (IR & ultrasonic) to determine the distance of the human presence from itself. It analyzes the gestures and responds accordingly by performing required operation. The robot can perform 3 such operations based on the users command.

Forward Motion :

This feature is used by the robot to move forward in its path. It can be enabled by moving the human presence (hand, palm or legs) backward and the robot will follow this gesture.

This feature uses the increasing distance between sensor and user to actuate the robot into moving.

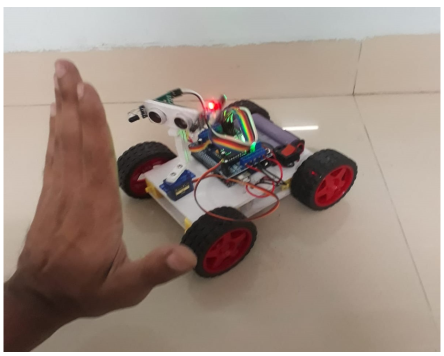

Fig. 1. Robot detecting presence of hand infront of it

The figure shows the detection of users hand by the sensors attached to robot. The "head" of the robot, where the sensor is attached which adjust according to the position of the user to get most appropriate detection range. The user will have to move the hand backward, away from the robot, to make it move forward.

Backward Motion :

This feature is used by the robot to move backward in its path. It works similar to the forward feature. The user should move closer to the robot and the sensor attached will detect the decreasing distance and actuate the robot into moving backward.

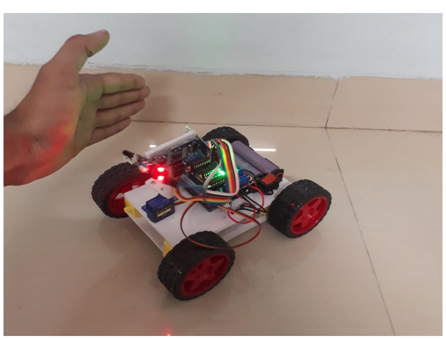

Fig. 2. Robot adjusting to detect hand

Standby Mode :

This is feature allows the robot to stop and wait for next instruction or command from user. Since there is no movement in the frame, the robot will stand by and wait for movement to occur.

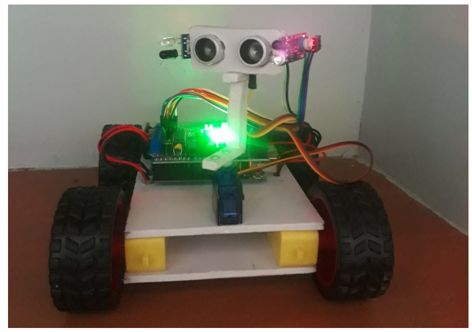

Fig. 3. Robot waiting for next instruction

All three operations can be understood using these steps:

1. User present in frame of sensor.
2. The motion of users hand or leg is detected by sensor.
3. Sensor sends this information to the Arduino UNO.
4. Arduino contains code with prefixed distance range. If the movement is in said range the system will detect it and send instructions to robot after analyzing it.
5. The 4 wheels and 4 motors attached which be actuated based on the instruction and move backward or forward depending on it.
6. If there is no movement in the frame, or if the presence is out of the sensors detection range, the robot receives instruction to wait for a command and goes into standby mode.

B. Testing

The project has been tested 3 times and has performed 3 operations namely, forward, backward, standby.

Below given are the requirements to run this project:

- Arduino Uno
- Arduino UNO IDE
- L293D Motor driver
- Infrared Sensors
- Ultrasonic Sensor
- Servo Motor
- Four DC Geared Motors
- Four Wheels
- Robot Chassis
- Jumper Wires
- 18650 Batteries
- Switch

C. Figures

Accuracy w.r.t distance of user from sensor graph

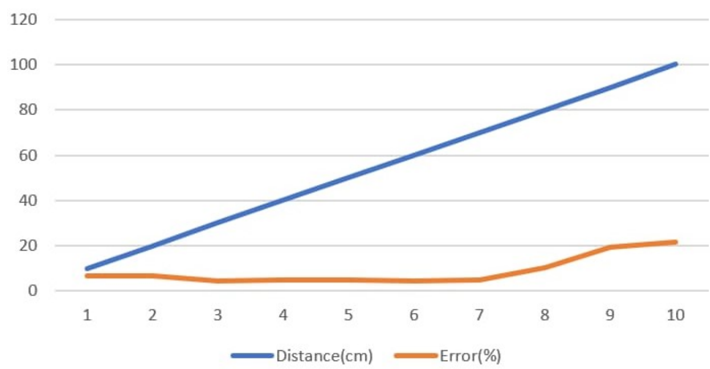

Fig. 4. Error v/s distance (in cm)

Figure 4 shows the graph of distance v/s error. The error % is plotted on the x – axis and distance (in cm) in plotted on the y – axis. The graph shows the most accurate range of distance is approximately 30cm to 70cm. In this range, the error % is lowest which indicate more accuracy

D. Image Files

Image of robot system

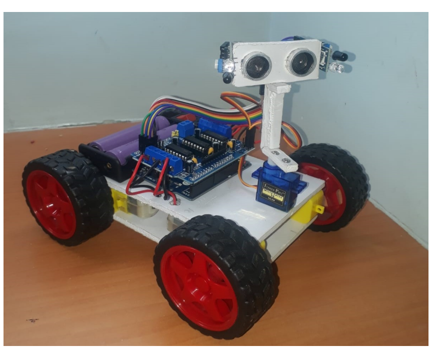

Fig. 5 Fully assembled, functioning robot system

Figure 5 shows the fully assembled and functioning robot. The ultrasonic sensor is mounted at a certain height to detect movement more efficiently. The arduino interface and connections are visible in the image. The 4 wheels attached are connected to 4 motor drivers which actuate motion in the robot according to instructions received.

IV. RESULTS AND DISCUSSIONS

The Project is designed using various hardware components such as motors, sensors, drivers etc. The basic design and idea of the system can be used as a blueprint for larger projects which use the same basic functionality and working of a robot. A wireless communication feature can be added in the robot to make it more versatile and control it from a large distance.

It can also be modified and used in a wide range of fields. A few examples are:

- To monitor surroundings by mounting a real time video recorder over the robot system.
- Provide public assistance in shopping malls, airports etc.
- Act as a luggage carrier and be used in medical and defence fields.
- This robot will be helpful to people with visual and speaking impairment as they can communicate with the robot using hand gestures easily and efficiently.

V. CONCLUSION

This project will be extremely useful in the coming years as the world shifts to a more technology dependent scenario. It was also provide assistance in eliminating work environments which are hazardous to humans. By modifying this robot, many more opportunities will be created for its use. Such as, handling and lifting heavy loads, performing repetitive tasks and also help prevent accidents at workplaces and factories.

VI. ACKNOWLEDGMENT

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